

» SEDIMENTARY ROCKS

**NUMMULITIC LIMESTONE**

A marine foraminiferan, *Nummulites* is the main fossil in this rock. The cement is calcite, originally lime mud.



*fossilized crinoid,
or sea lily, stalk*

CRINOIDAL LIMESTONE

Crinoids are echinoderms that are attached to the seabed by a flexible stalk. Crinoidal limestone is a mass of broken stalks cemented by hardened lime mud.

**FRESHWATER LIMESTONE**

This limestone is a pale, calcite-rich rock with some quartz and clay. It contains fossils of freshwater organisms, which indicate the environment of deposition.



*pisolith cemented
by calcite*

PISOLITIC LIMESTONE

This rock is composed of pisoliths – pea-sized grains slightly larger than oolites, often flattened, and loosely cemented by calcite.

OOOLITIC LIMESTONE

This limestone is composed of oolites – small, rounded, concentrically banded grains rolled by seabed currents and cemented by carbonate mud.

CORAL LIMESTONE

This rock is a mass of fossilized corals cemented by fine-grained calcite. It is grey to white or brownish in colour.

**BRYOZOAN LIMESTONE**

This is a grey or reddish organic limestone, which contains fossils of bryozoans in a matrix of hardened, calcite-rich mud.

**LIMESTONE BRECCIA**

Often forming below limestone cliffs, this breccia comprises large, angular fragments of limestone and other rocks, cemented by calcite.

**FELDSPATHIC GRITSTONE**

Coarse-grained and pale- to dark-coloured, this gritstone contains much quartz and up to 25 per cent feldspar.

**QUARTZ GRITSTONE**

This gritstone is composed of quartz with some feldspar and mica, all of coarse grain size.

**GREYWACKE**

This dark rock contains quartz, rock fragments, and feldspars, set in a mass of finer clay and chlorite. It forms in marine basins.

**ARKOSE**

Variable in colour and medium-grained, arkose is a sandstone with a high percentage of feldspar.

**DOLOMITE**

This yellowish brown or grey rock contains a high percentage of dolomite (calcium magnesium carbonate). It is also called dolostone, to distinguish it from this mineral.

FOSSILIFEROUS SHALE

Fine-grained marine sedimentary rocks such as shale often contain large numbers of well-preserved fossils.

SHALE

This fine-grained, layered rock varies in composition, usually containing silt, clay minerals, organic materials, iron oxides, and minute crystals of minerals such as pyrite and gypsum.



brachiopod fossil